FACULTY OF COMPUTING

SEMESTER 1 2025/2026

SECJ3553 – ARTIFICIAL INTELLIGENCE

SECTION 3

ASSIGNMENT 2: STATE SPACE SEARCH

THEME: SMART CITY

LECTURER: DR. NORSHAM BINTI IDRIS

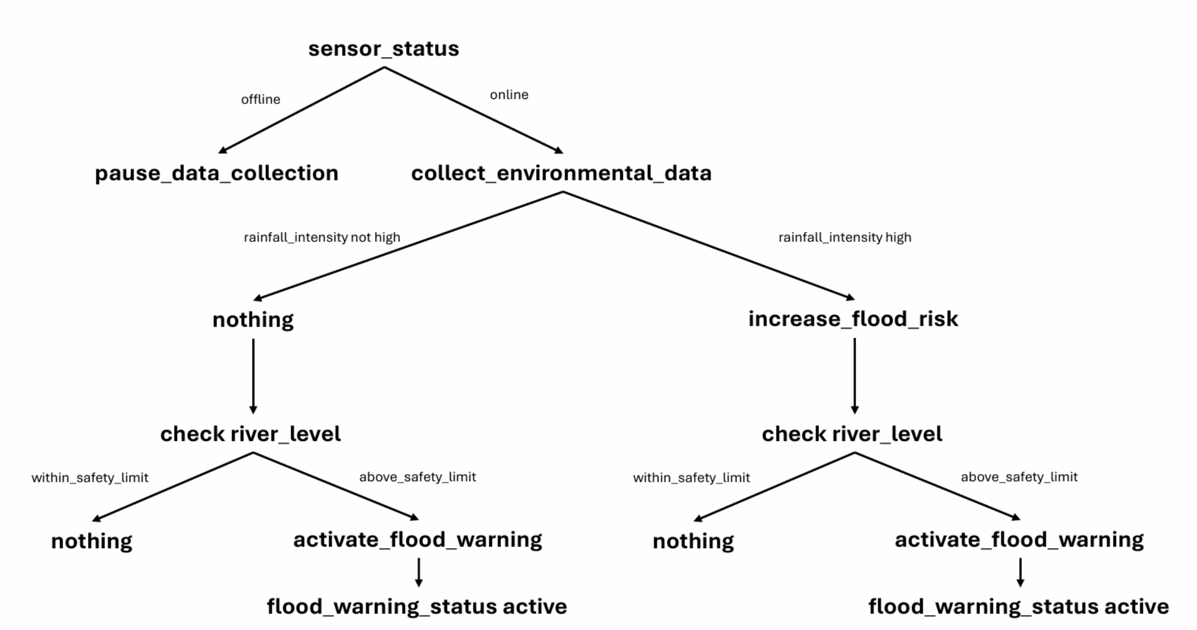
GROUP 7

NAME OF STUDENTS	MATRIC NO.
TAN ZHI MING	A23CS0189
CHUA JIA LIN	A23CS0069
NAZMI HAIKAL BIN KHAIRUL	A23CS0145
POH LOK YEE	A23CS0262

Table of Contents

Problem Formulation by State Space Graph

1.0 Environmental Data Collection (IoT Sensor Layer)



1.1 Explanation States and Actions

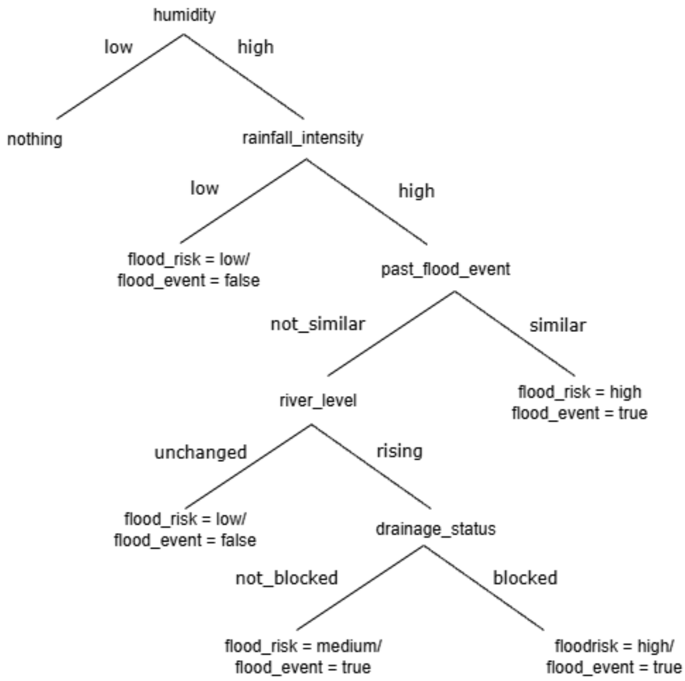
States	a. **rainfall_intensity high:** a state where the rainfall sensor reports that the rainfall intensity is above the predefined safety threshold. b. **river_level above_safety_limit:** a state where the river-level sensor detects that the water level is higher than the configured safety limit. c. **sensor_status offline:** a state where one or more sensors (rainfall, humidity, river level) stop sending data or lose connection. d. **sensor_status online:** a state where all required sensors are active and continuously sending valid readings. e. **flood_warning_status active:** a state where the system has activated flood warning due to high river level and/or high rainfall intensity. f. **flood_risk high:** a state where the current combination of sensor readings indicates an increased risk of flooding.
Actions	a. **collect_environmental_data:** action where online sensors periodically capture rainfall, humidity, and river-level readings and send them to the system. b. **increase_flood_risk:** action that updates the internal flood_risk state to high when rainfall_intensity is high or other conditions indicate danger. c. **activate_flood_warning:** action that sets flood_warning_status to active once river_level is above the safety_limit. d. **pause_data_collection:** action that pauses or flags data collection when sensor_status becomes offline so that invalid or missing data is not used by the AI model.

	e. **nothing:** action where no change is made to the system when all environmental variables remain within safe limits.

1.2 Problem Formulation and Explanation

Critical	Details	Explanation
Initial State	sensor_status online	The system will first detect whether it is raining at the moment. If it is raining, then the system will check the drainage condition.
Action	Let S(x) = Set of action–state pairs. S(sensor_status) = {offline: pause_data_collection, online: collect_environmental_data} S(rainfall_intensity) = {high: increase_flood_risk, not_high: nothing} S(river_level) = {above_safety_limit: activate_flood_warning, within_safety_limit: nothing}	The system manages environmental sensing and flood indicators through these action–state pairs. When sensors are online, they continuously collect data. If rainfall_intensity is high, the system increases the flood_risk state. If river_level goes above the safety limit, the system activates flood_warning_status. If any sensor becomes offline, data_collection is paused or flagged so that unreliable readings are not used by the prediction model.
Goal	flood_warning_status active	The goal of the state space representation is to reach the state where the system activates the flood warning. This happens when the environmental conditions such as high rainfall intensity and river level above the safety limit — trigger the system to set the flood warning status to active.
Path Cost	Cost on the route: sensor_status online → collect_environmental_data → evaluate rainfall_intensity and river_level → update flood_risk / flood_warning_status.	In this scenario, the path cost represents the time and resources required for each sensing–evaluation–action cycle (sampling interval, network latency, processing time, and energy usage). A lower path cost means the system can update flood_risk and flood_warning_status more quickly, improving the timeliness and reliability of early flood warnings.

2.0 Flood Prediction Model



2.1 Explanation States and Actions

States	a. **Humidity:** a state where system checks the humidity level (high or low) b. **rainfall_intensity:** an action where system checks the rainfall intensity (moderate or high). c. **past_flood_event:** an action where system checks the historical pattern (not_similar or similar). d. **river_level:** an action where system checks the river condition (unchanged or rising). e. **drainage_status:** an action where system checks the drainage condition (not_blocked or blocked).
Actions	a. **flood_risk = Low:** an action that updates the internal flood risk to a low level. b. **flood_risk = High:** an action that updates the internal flood risk to a high level. c. **flood_event = True:** an action that the internal prediction that a flood event will occur when certain conditions are meet. d. **flood_event = False:** an action that the internal prediction that a flood event will not occur. e. **nothing:** action where no change and system continue monitoring.

2.2 Problem Formulation and Explanation

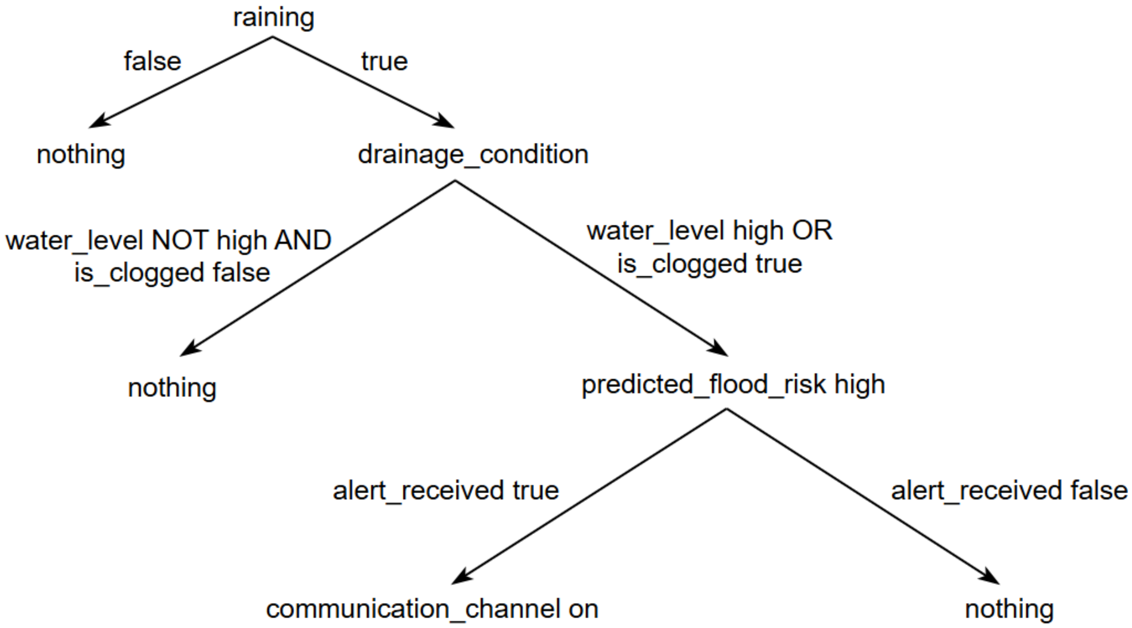
Critical	Details	Explanation
Initial State	humidity	The process begins by checking the humidity level. This is the initial fact set that will trigger further reasoning. The system must transition to high humidity branch to proceed, or it will continue monitoring the status.
Action	Let S(x) = Set of action–state pairs. S(rainfall_intensity) = {low: flood_risk low and flood_event false, high: past_flood_event} S(past_flood_event) = {similar: flood_risk high and flood_event true, not_similar: river_level} S(river_level) = {rising: drainage_status, unchanged: flood_risk low and flood_event false} S(drainage_status) = {blocked: flood_risk high and flood_event true, not_blocked: flood_risk medium and flood_event true}	The system manages flood risk assessment through a sequential set of action-state pairs. It predicts the flood risk based on the measured rainfall_intensity and subsequent detailed environmental checks. If the rainfall_intensity is low, the system will conclude there is low flood risk and flood_event is false. When rainfall is high, the system gets cautions and check the past flood event. If the heavy rain matches the past_flood_event, the system immediately flags high risks, and the flood event may occur. If the past records are clear, the system will then look at the river_level. If the river_level is rising, the system will check the drainage_status. If not, then the system will flag low risks, and the flood event will not occur. If the drains are working, the danger is lowered to medium risk, otherwise the risk will be high, and flood may occur soon. This process makes sure the system is quick, clear, and only gives a top warning when conditions are truly severe.

Goal	flood_event = true	The goal is to successfully achieve the prediction the occurrence of flood event. In other words, after utilizing the chain of environmental and historical data checks, the system can decide if a flood is coming. Achieving this goal ensures timely warnings so authorities and residents can take preventive actions and reduce potential damage.
Path Cost	Cost on the route: 1) humidity→ rainfall_intensity→ past_flood_event→ flood_event 2) humidity→ rainfall_intensity→ past_flood_event→ river_event→ drainage_status→ flood_event	In this scenario, the path cost represents the time and effort of the system takes to reach a flood prediction event with true. Route 2 requires more checks, like evaluating river levels and drainage, which naturally take longer, while Route 1 reach a prediction more quickly. This ensures the system adapts its process based on the conditions it observes.

Solution:

1) humidity, rainfall_intensity, past_flood_event, flood_event
2) humidity, rainfall_intensity, past_flood_event, river_event, drainage_status, flood_event

3.0 Alert and Communication System



3.1 Explanation States and Actions

States	a. **raining:** a state where it is raining at that moment b. **nothing:** a state where nothing is happening c. **drainage_condition:** the state where it will evaluate whether the water level is high or not, and whether the drainage is clogged or not when it is raining d. **predicted_flood_risk high:** a state where the flood risk is predicted as high when the water level is high or the drainage is clogged, and it is raining at that time e. **communication_channel on:** a state where the communication channel is turned on when residents, city planners, or emergency teams received the flood alert
Actions	a. **false:** action for state checking condition to be false b. **true:** action for state checking condition to be true c. **water_level NOT high AND is_clogged false**: action that states the water level is not high and the drainage is not clogged d. **water_level high OR is_clogged true:** action that states the water level is high and the drainage is clogged e. **alert_received true:** action that states the flood alert is received by residents, city planners, or emergency teams f. **alert_received false:** action that states the flood alert is not received by residents, city planners, or emergency teams

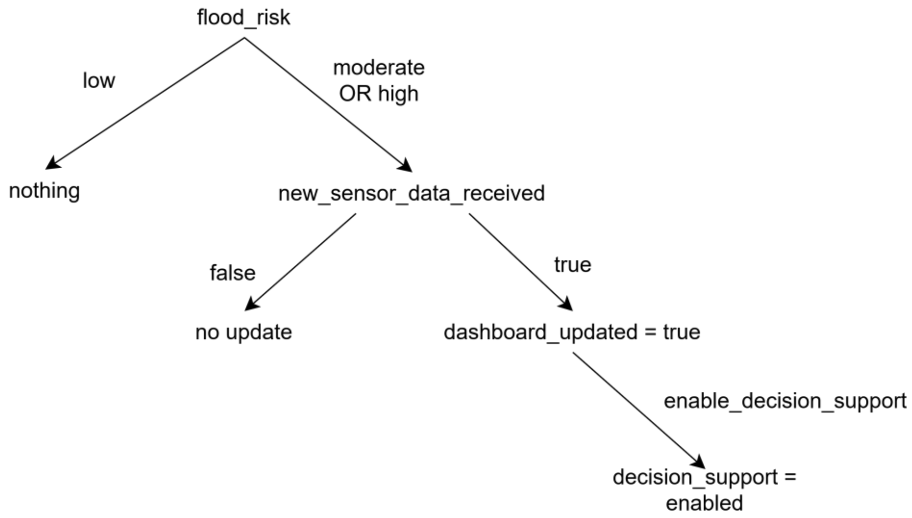
3.2 Problem Formulation and Explanation

Critical	Details	Explanation
Initial State	raining	The system will first detect whether it is raining at the moment. If it is raining, then the system will check the drainage condition.
Action	Let S(x) = Set of action-state pairs. S(raining) = {false: nothing, true: drainage_condition} S(drainage_condition) = {water_level NOT high AND is_clogged false: nothing, water_level high OR is_clogged true: predicted_flood_risk high} S(predicted_flood_risk high) = {alert_received true: communication_channel on, alert_received false: nothing}	The system manages raining and drainage condition through a set of action-state pairs. It predicts the flood risk based on information such as the drainage condition and is it raining at that moment. If it is raining, and the water level is high or the drainage is clogged, then an alert will be sent to residents, city planners, and emergency teams. If anyone received the alert, a communication channel will be turned on to allow real-time communication. If these conditions are not satisfied, then no actions will be taken.
Goal	communication_channel on	The goal that we want to achieve here is to open a real-time communication channel for residents, city planners, and emergency teams to speed up the rescue actions and perform preventive measures when high flood risk is predicted.
Path Cost	Cost on the route: raining → drainage_condition → predicted_flood_risk high → communication_channel on	In this scenario, the path cost is the execution time taken for the system to reach the goal which is to open a real-time communication channel for residents, city planners, and emergency teams.

Solution:

1) raining, drainage_condition, predicted_flood_risk high, communication_channel on

4.0 Geographic Information System and Decision Support System



4.1 Explanation States and Actions

States	a. **flood_risk high**: a state where analytics indicate severe flood danger based on sensor readings and prediction models. b. **flood_risk moderate**: a state where the flood probability is noticeable but not critical. c. **flood_risk low**: a state where flood likelihood is minimal. d. **new_sensor_data_received true**: a state where new readings (rainfall, river level, drainage condition, humidity) are received by the system. e. **dashboard_updated true**: a state where the GIS dashboard has refreshed its map layers and indicators. f. **decision_support enabled**: a state where decision-making assistance (recommendations, warnings, evacuation planning) is activated. g. **map_indicator red_zone**: a state showing high-risk areas in red on the GIS map. h. **map_indicator yellow_zone**: a state showing moderate-risk zones in yellow. i. **map_indicator green_zone**: a state showing safe or low-risk locations in green.
Actions	a. **update_map_indicator**: action that updates the GIS map color (red, yellow, green) based on flood risk levels. b. **update_dashboard**: action that refreshes the dashboard when new sensor data arrives. c. **enable_decision_support**: action that activates routing suggestions, risk analysis, and response recommendations. d. **nothing**: action where no update is required if data remains unchanged.

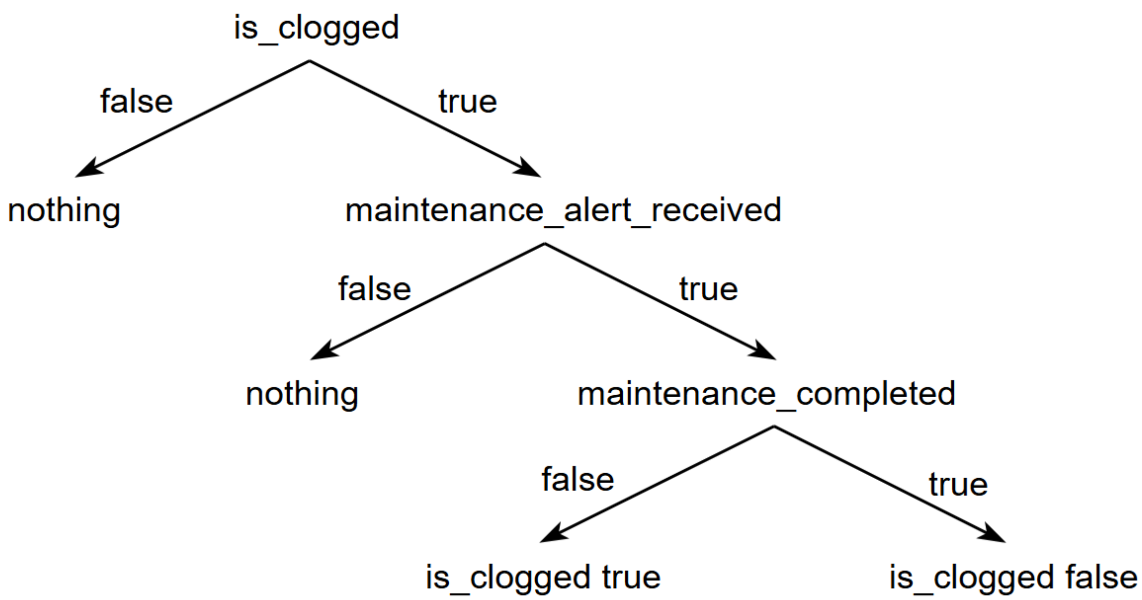
4.2 Problem Formulation and Explanation

Critical	Details	Explanation
Initial State	flood_risk low	The system begins at a normal condition where all monitored areas are considered low risk. GIS displays all regions marked as safe (green zone). The dashboard remains stable until new sensor data triggers updates.
Action	Let S(x) = Set of action-state pair S(flood_risk) = { High:map_indicator red_zone, Moderate: map_indicator yellow_zone, Low: map_indicator green_zone} S(new_sensor_data_received) = {true: update_dashboard, false: nothing} S(dashboard_updated) = {true: decision_support enabled, false: nothing}	Through these action–state pairs, GIS visualizes real-time flood conditions by updating map indicators based on risk levels. When new sensor readings are received, the dashboard refreshes automatically. Once updated, the system activates a decision support layer that assists residents and authorities with planning, evacuation routes, and resource allocation.
Goal	decision_support enabled	The intended goal of this subsystem is to activate a fully functional decision support system that helps emergency teams, residents, and city planners take timely actions. By integrating sensor data and flood predictions, the GIS ensures critical information is presented clearly in an actionable manner.
Path Cost	Cost on the route: flood_risk → new_sensor_data_received → update_dashboard → decision_support enabled	The path cost represents the computational, rendering, and communication time required for the system to update all GIS layers and enable decision support functions. Lower path cost ensures faster visualization and sharper response time during emergencies, improving safety and reducing potential damage.

Solution:

flood_risk, new_sensor_data_received, update_dashboard, dashboard_updated, decision_support enabled

5.0 Drainage Situation Analysis and Management



5.1 Explanation States and Actions

States	a. **is_clogged:** a state where the drainage is clogged at that moment b. **nothing:** a state where nothing is happening c. **maintenance_alert_received:** the state where it will evaluate whether the city planners received the maintenance alert or not d. **maintenance_completed:** a state where it will evaluate whether the drainage maintenance is done is not e. **is_clogged true**: a state where the drainage is clogged f. **is_clogged false:** a state where the drainage is not clogged
Actions	a. **false:** action for state checking condition to be false b. **true:** action for state checking condition to be true

5.2 Problem Formulation and Explanation

Critical	Details	Explanation
Initial State	is_clogged	The system will first detect whether the drainage is clogged or not. If it is clogged, then the system will send a maintenance alert to city planners.

Action	Let S(x) = Set of action-state pairs. S(is_clogged) = {false: nothing, true: maintenance_alert_received} S(maintenance_alert_received) = {false: nothing, true: maintenance_completed} S(maintenance_completed) = {false: is_clogged true, true: is_clogged false}	The system manages is_clogged through a set of action-state pairs. It predicts drainage maintenance needs based on drainage cloggage. If the drainage is clogged, then an alert will be sent to city planners. If city planners received the maintenance alert, the drainage maintenance will be carried out to solve the cloggage. If the drainage is not clogged, then no actions will be taken.
Goal	is_clogged false	The goal that we want to achieve here is to solve the cloggage and change the is_clogged status to false after maintenance is completed.
Path Cost	Cost on the route: is_clogged → maintenance_alert_received → maintenance_completed → is_clogged false	In this scenario, the path cost is the execution time taken for the system to reach the goal which is to change the is_clogged status to false after maintenance is completed.

Solution:

1) is_clogged, maintenance_alert_received, maintenance_completed, is_clogged false